

CS 486/686

Probabilities

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Lecture 6

RN 12.2–12.3 · PM 8.1

Reminders

DUE TONIGHT

Thu May 28, 11:59 PM

Chat 2 (Heuristic search & CSPs) on Chrysalis.

OUT TODAY

Due Tue Jun 2

Chat 3 (Probabilities) — released today.

Search ›

Uncertainty ›

Decisions ›

Learning

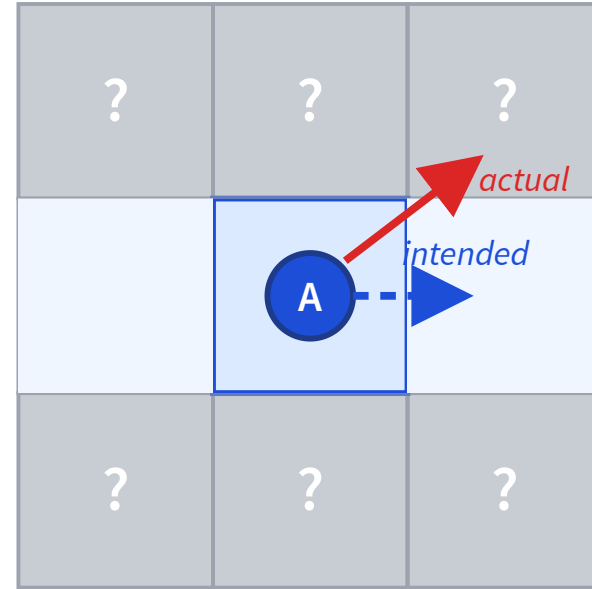
Learning goals

Calculate prior, posterior, and joint probabilities using:

- the **sum rule** (marginalization),
- the **product rule** (conditioning),
- the **chain rule** (joint factorization),
- and **Bayes' rule** (flipping a conditional).
- Then put it all together with a **universal 3-step recipe**.

Why handle uncertainty?

- Can't see the whole state — partial observation.
- Actions may not produce their intended outcome.
- **Reason about uncertainty** and decide anyway.



Visible cells are clear; the rest are hidden.
Actions drift.

Frequentists vs Bayesians

Two camps for the formal measure of uncertainty.

Frequentist

Long-run frequency. *Objective*. After n

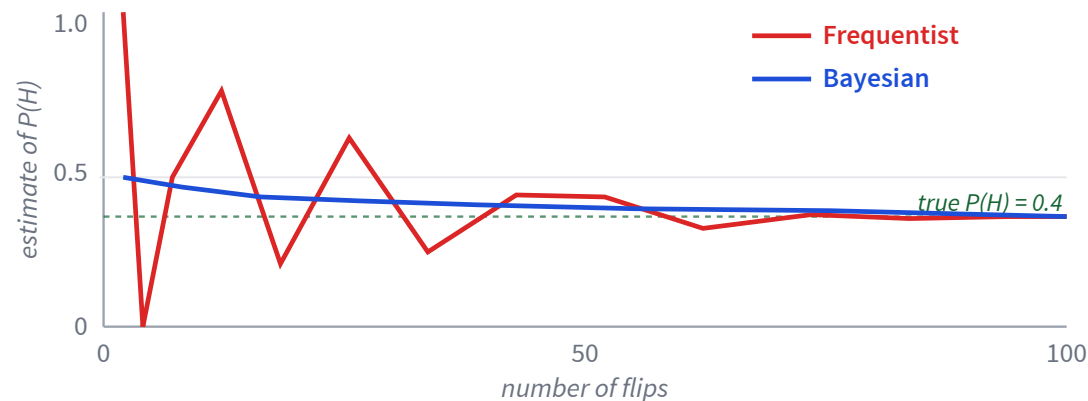
$$H, m T: P(H) = \frac{n}{n+m}. \text{ Needs data.}$$

Bayesian

Degree of belief: prior + evidence.

$$\text{Laplace 1H/1T: } P(H) = \frac{1+n}{2+n+m}.$$

Works with no data.

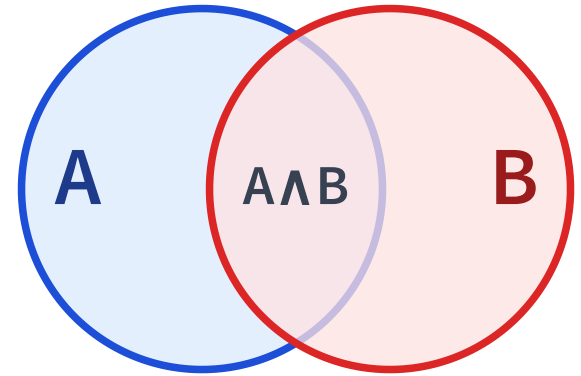


Random variables

- A *domain* of values + a *distribution* over them.
- e.g. A = “alarm sounding”: domain $\{T, F\}$, $P(A=T) = 0.1$, $P(A=F) = 0.9$.
- Boolean shorthand: $P(A) \equiv P(A=T)$; $P(\neg A) \equiv P(A=F)$.

Axioms of probability

- $0 \leq P(A) \leq 1$
- $P(\text{true}) = 1$; $P(\text{false}) = 0$
- **Inclusion-exclusion:** $P(A \vee B) = P(A) + P(B) - P(A \wedge B)$



$0 < P(A) < 1$ means we're *uncertain*, not that A is partly true.

Joint probability distribution

- **Atomic event:** an assignment to every variable.
- **Joint distribution:** a probability for every atomic event.

e.g. $P(\text{weather, temperature})$

	Hot	Mild	Cold
Sunny	0.10	0.20	0.10
Cloudy	0.05	0.35	0.20

Probabilities sum to 1 across the table.

Prior vs posterior

- Prior $P(X)$ – belief in X with no other info.
- Posterior $P(X \mid Y)$ – belief in X given evidence Y .



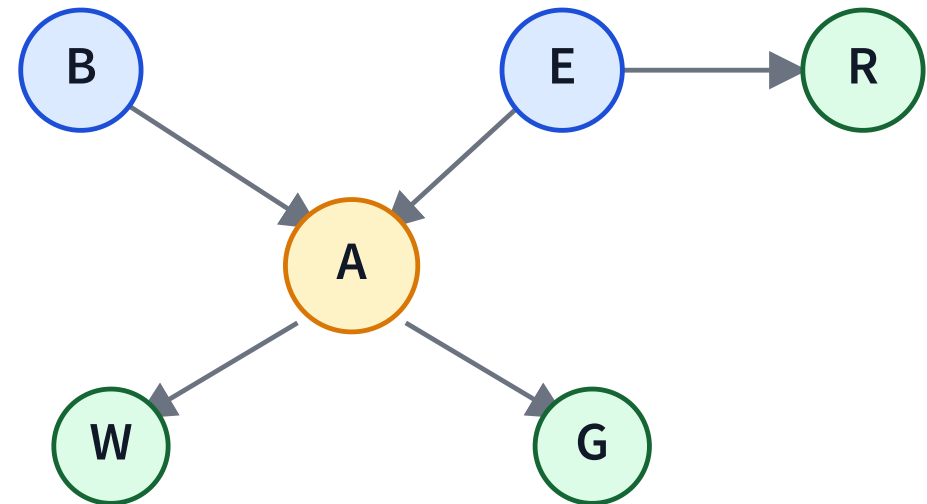
Running example: Holmes' burglar alarm

Mr. Holmes has a burglar alarm and two flaky neighbours (Watson, Gibbon) who call when they hear it. The alarm also misfires on earthquakes; earthquakes are reported on the radio.

6 Boolean variables $\Rightarrow 2^6 = 64$ joint probabilities.

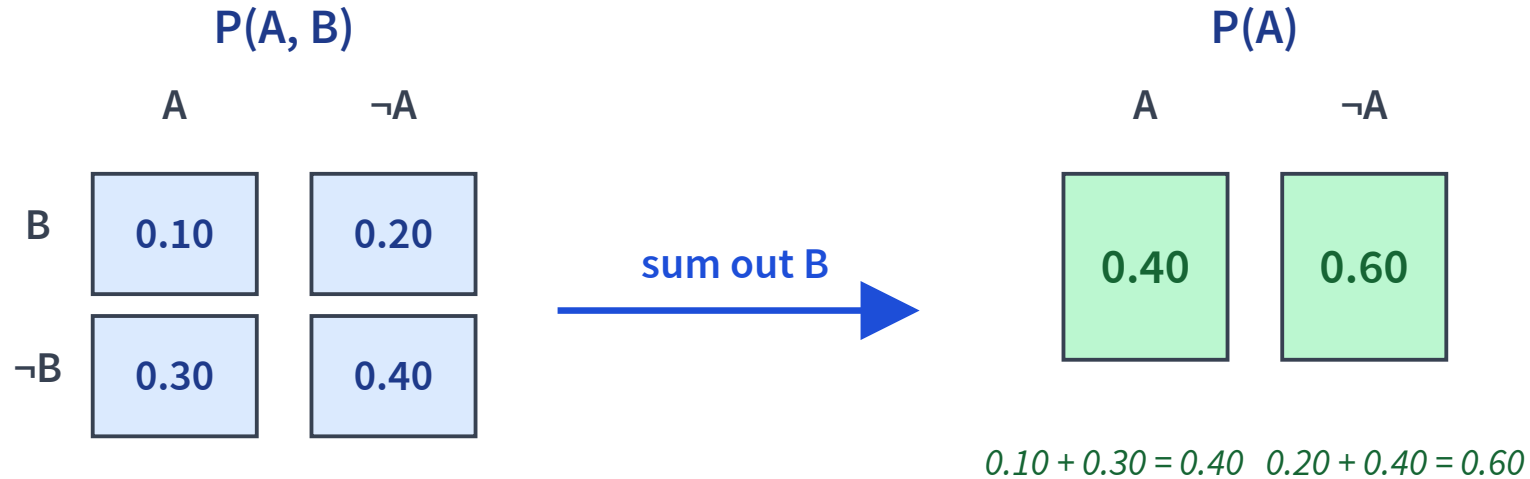
Random variables

- B : burglary
- E : earthquake
- A : alarm sounding
- W : Watson calling
- G : Gibbon calling



The sum rule (marginalization)

From a joint, get a probability over a *subset* by summing out the rest.



In general: $P(A) = \sum_b P(A \wedge B=b)$

Marginalizing the Holmes joint

$P(A, W, G)$ over Watson W , Gibbon G , and the alarm A :

	A		$\neg A$	
	G	$\neg G$	G	$\neg G$
W	0.032	0.048	0.036	0.324
$\neg W$	0.008	0.012	0.054	0.486

Example — $P(\neg A \wedge W)$: sum out G within the highlighted cells.

$$P(\neg A \wedge W) = 0.036 + 0.324 = \mathbf{0.36}$$

Quiz pack: marginalization

Q1. $P(\neg A \wedge W)$?

$$= 0.036 + 0.324 = \mathbf{0.36}.$$

Q2. $P(A \wedge \neg G)$?

$$= 0.048 + 0.012 = \mathbf{0.06}.$$

Q3. $P(\neg A)$?

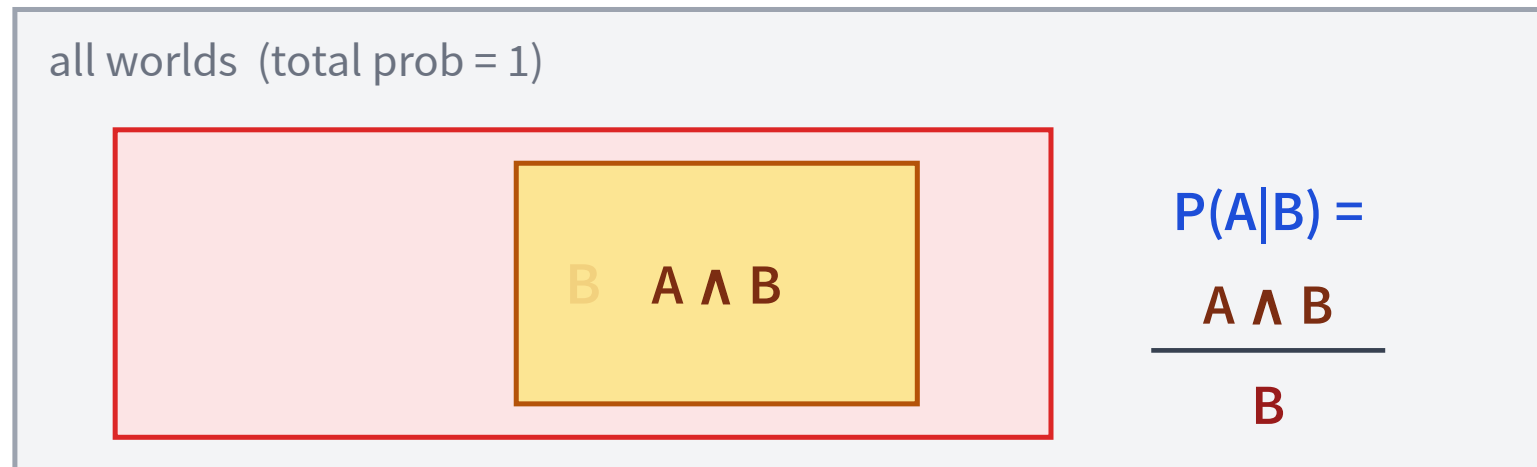
$$= 0.036 + 0.324 + 0.054 + 0.486 = \mathbf{0.9}.$$

	A		$\neg A$	
	G	$\neg G$	G	$\neg G$
W	0.032	0.048	0.036	0.324
$\neg W$	0.008	0.012	0.054	0.486

Conditional probability and the product rule

$P(A|B)$ = fraction of the B -world where A also holds.

$$P(A|B) = \frac{P(A \wedge B)}{P(B)}$$



Multiply both sides by $P(B)$:

$$P(A \wedge B) = P(A|B) P(B)$$

$$\text{Q: } P(W \mid \neg A)?$$

From the previous slides: $P(\neg A \wedge W) = 0.36$, $P(\neg A) = 0.9$.

Q. Watson calls, given the alarm is NOT going.

A. 0.2

B. 0.4

C. 0.6

D. 0.8

$$\text{B. } P(W \mid \neg A) = \frac{P(\neg A \wedge W)}{P(\neg A)} = \frac{0.36}{0.9} = \mathbf{0.4}.$$

$$\text{Q: } P(\neg G \mid A)?$$

From the previous slides: $P(A \wedge \neg G) = 0.06$, $P(\neg A) = 0.9$ (so $P(A) = 0.1$).

Q. Gibbon does NOT call, given the alarm is going.

A. 0.2

B. 0.4

C. 0.6

D. 0.8

$$\text{c. } P(\neg G \mid A) = \frac{P(A \wedge \neg G)}{P(A)} = \frac{0.06}{0.1} = \mathbf{0.6}.$$

The chain rule (joint factorization)

The product rule, repeated.

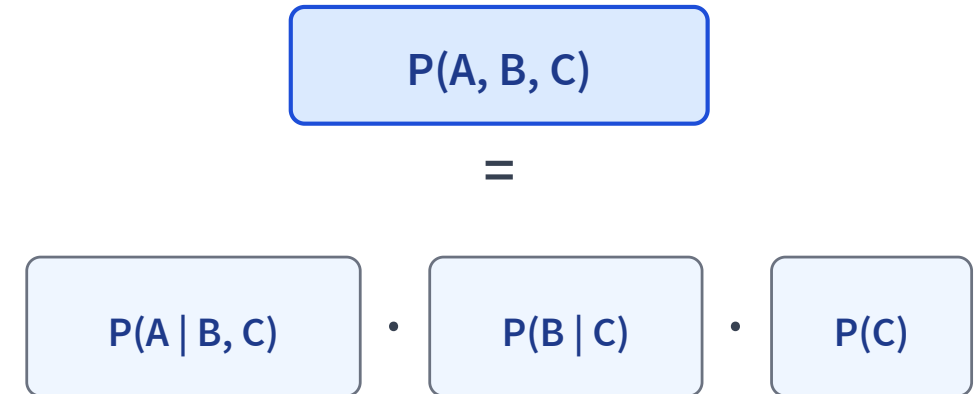
2 vars: $P(A, B) = P(A|B) P(B)$

3 vars:

$$P(A, B, C) = P(A|B, C) P(B|C) P(C)$$

In general:

$$P(X_1, \dots, X_n) = \prod_{i=1}^n P(X_i | X_1, \dots, X_{i-1})$$



*condition each on its predecessors
(any ordering works)*

$$\text{Q: } P(A \wedge W \wedge \neg G)?$$

Given: $P(A) = 0.1$, $P(W|A) = 0.9$, $P(G|A \wedge W) = 0.3$.

Q. Apply the chain rule to the conjunction.

A. 0.027

B. 0.037

C. 0.054

D. 0.063

$$\text{D. } P(A) \cdot P(W|A) \cdot P(\neg G|A \wedge W) = 0.1 \times 0.9 \times 0.7 = \mathbf{0.063}.$$

$$\mathbf{Q: } P(\neg A \wedge \neg W \wedge \neg G)?$$

Given: $P(A) = 0.1$, $P(W|\neg A) = 0.4$, $P(G|\neg A \wedge \neg W) = 0.1$.

Q. Apply the chain rule, this time to the all-negated conjunction.

A. 0.486

B. 0.324

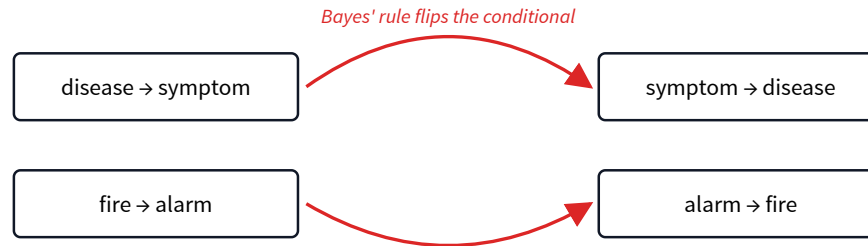
C. 0.216

D. 0.054

$$\mathbf{A. } P(\neg A) \cdot P(\neg W|\neg A) \cdot P(\neg G|\neg A \wedge \neg W) = 0.9 \times 0.6 \times 0.9 = \mathbf{0.486}.$$

Why flip conditionals?

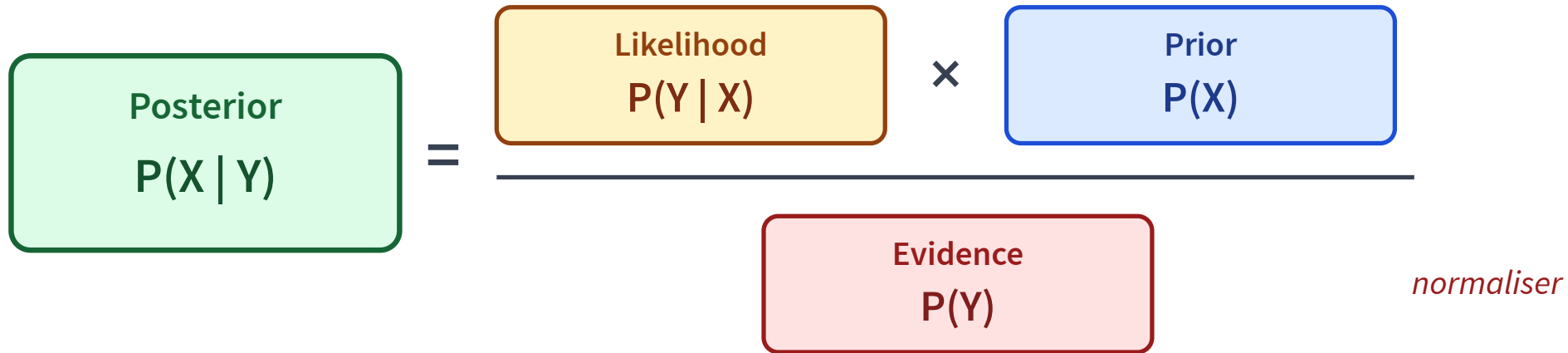
Models give us **causal** knowledge. Diagnosis needs **evidential** reasoning.



We know $P(\text{symptom}|\text{disease})$ and want $P(\text{disease}|\text{symptom})$.

Bayes' rule

Bayes' rule. $P(X|Y) = \frac{P(Y|X) P(X)}{P(Y)}$



A diagram illustrating the components of Bayes' rule. On the left, a green rounded rectangle contains the text "Posterior" and $P(X | Y)$. To its right is an equals sign. Further right is a fraction. The numerator consists of two rounded rectangles: a yellow one labeled "Likelihood" and $P(Y | X)$, followed by a multiplication symbol $\times$, and then a blue one labeled "Prior" and $P(X)$. A horizontal line separates the numerator from the denominator. The denominator is a red rounded rectangle labeled "Evidence" and $P(Y)$. To the right of the denominator box, the word "normaliser" is written in red italicized font.

$$\text{Posterior } P(X | Y) = \frac{\text{Likelihood } P(Y | X) \times \text{Prior } P(X)}{\text{Evidence } P(Y) \text{ (normaliser)}}$$

Derive it from the product rule: $P(X, Y) = P(X|Y) P(Y) = P(Y|X) P(X)$
 $\rightarrow$ divide by $P(Y)$.

$$\text{Q: } P(\neg A \mid W)?$$

Given: $P(A) = 0.1$, $P(W|A) = 0.9$, $P(W|\neg A) = 0.4$.

Q. Alarm NOT going, given Watson calls.

- A. 0.2
- B. 0.45
- C. 0.8
- D. 0.9

C. Marginalize: $P(W) = P(A)P(W|A) + P(\neg A)P(W|\neg A) = 0.09 + 0.36 = 0.45$.

$$\text{Bayes: } P(\neg A|W) = \frac{P(\neg A)P(W|\neg A)}{P(W)} = \frac{0.36}{0.45} = \mathbf{0.8}.$$

$$\text{Q: } P(A \mid \neg G)?$$

Given: $P(A) = 0.1, P(G|A) = 0.3, P(G|\neg A) = 0.1$.

Q. Alarm going, given Gibbon does NOT call.

A. 0.03

B. 0.08

C. 0.30

D. 0.70

B. Marginalize: $P(\neg G) = P(A)P(\neg G|A) + P(\neg A)P(\neg G|\neg A) = 0.07 + 0.81 = 0.88$.

$$\text{Bayes: } P(A|\neg G) = \frac{P(A)P(\neg G|A)}{P(\neg G)} = \frac{0.07}{0.88} \approx \mathbf{0.08}.$$

A universal approach

Three rules — one per kind of probability you need.

1. **To calculate a conditional probability** — write it as a ratio of two joint probabilities (*product rule, reversed*):
$$P(X|Y) = \frac{P(X \wedge Y)}{P(Y)}.$$
2. **To calculate a partial joint** (some variables missing) — sum over the missing variables to get full joints (*sum rule, reversed*).
3. **To calculate a full joint** (all variables) — factor as a product of conditionals (*chain rule*).

Worked example: $P(A|C)$

Given:

$$P(A) = 0.6$$

$$P(B|A) = 0.4, P(\neg B|\neg A) = 0.2$$

$$P(C|A, B) = 0.1, P(C|\neg A, B) = 0.2, P(C|A, \neg B) = 0.5, P(C|\neg A, \neg B) = 0.8$$

Step 1. Convert the conditional into a ratio of joints:

$$P(A|C) = \frac{P(A \wedge C)}{P(C)} = \frac{P(A \wedge C)}{P(A \wedge C) + P(\neg A \wedge C)}$$

Worked example: step 2 (marginalize)

From step 1: $P(A|C) = \frac{P(A \wedge C)}{P(A \wedge C) + P(\neg A \wedge C)}$. The numerator and denominator are partial joints — sum out the missing variable B .

Step 2. Sum out B :

$$\begin{aligned}P(A \wedge C) &= P(A \wedge B \wedge C) + P(A \wedge \neg B \wedge C) \\P(\neg A \wedge C) &= P(\neg A \wedge B \wedge C) + P(\neg A \wedge \neg B \wedge C)\end{aligned}$$

Now every term on the right is a full-joint probability, ready for the chain rule.

Worked example: step 3 + answer

Apply the chain rule $P(A \wedge B \wedge C) = P(C|A \wedge B) P(B|A) P(A)$ to each full joint.

$$P(A \wedge B \wedge C) = 0.1 \times 0.4 \times 0.6 = 0.024$$

$$P(A \wedge \neg B \wedge C) = 0.5 \times 0.6 \times 0.6 = 0.180$$

$$P(\neg A \wedge B \wedge C) = 0.2 \times 0.8 \times 0.4 = 0.064$$

$$P(\neg A \wedge \neg B \wedge C) = 0.8 \times 0.2 \times 0.4 = 0.064$$

$$P(A|C) = \frac{0.024 + 0.180}{0.024 + 0.180 + 0.064 + 0.064} = \frac{0.204}{0.332} \approx \mathbf{0.614}$$

Four rules at a glance

Rule	Formula	Use when
Sum	$P(A) = \sum_b P(A \wedge B=b)$	marginalizing out variables
Product	$P(A \wedge B) = P(A B) P(B)$	defining or applying a conditional
Chain	$P(X_1 \wedge \dots \wedge X_n) = \prod_i P(X_i X_1 \wedge \dots \wedge X_{i-1})$	computing a full joint from conditionals
Bayes'	$P(X Y) = \frac{P(Y X) P(X)}{P(Y)}$	flipping causal $\rightarrow$ evidential

Learning goals (recap)

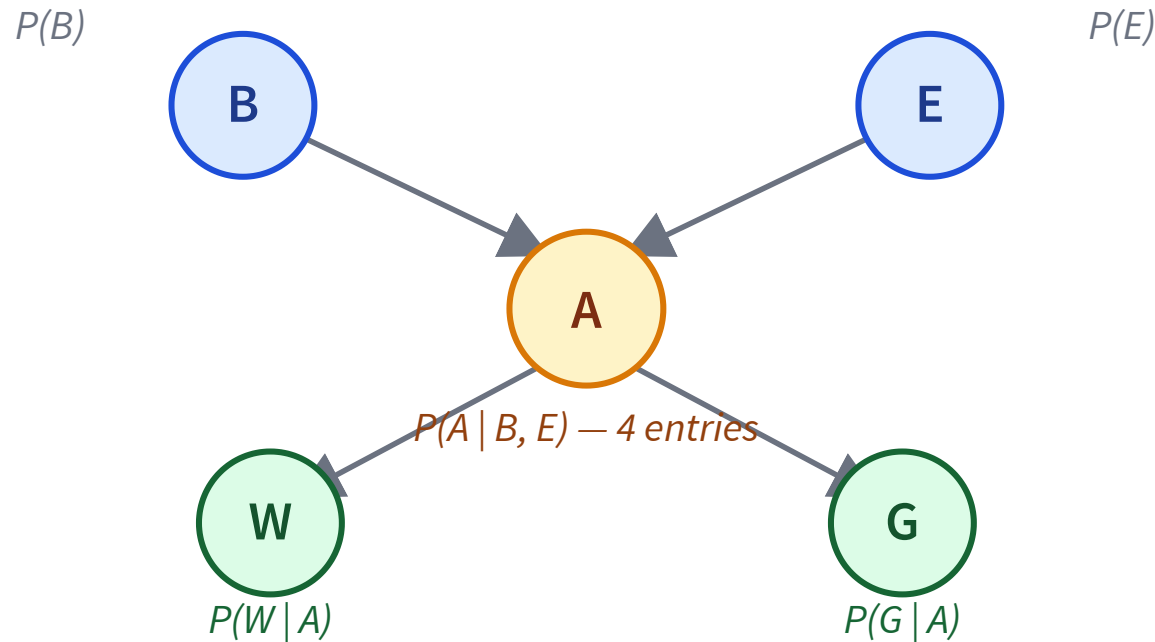
Calculate prior, posterior, and joint probabilities using:

- ✓ the **sum rule**
- ✓ the **product rule**
- ✓ the **chain rule**
- ✓ **Bayes' rule**
- ✓ the **universal 3-step recipe**

Next: independence and Bayes nets

Chain rule gives full joints — but 2^n numbers is too many.

Conditional independence + Bayesian networks → far fewer numbers.



5 variables: 32 joint numbers → ~10 CPT entries.